# In Catilinam IV ⋆

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**Abstract**

Cum M. Cicero consul Nonis Decembribus senatum in aede Iovis Statoris consuleret, quid de iis coniurationis Catilinae sociis fieri placeret, qui in custodiam traditi essent, factum est, ut duae potissimum sententiae proponerentur, una D. Silani consulis designati, qui morte multandos illos censebat, altera C. Caesaris, qui illos publicatis bonis per municipia Italiae distribuendos ac vinculis sempiternis tenendos existimabat.

*Key words:* Cicero; Catiline; orations.

## 1 Introduction

Gauge invariance is a central organizing principle of modern physics, expressing the idea that observable quantities should remain unchanged under certain local transformations of the mathematical description. In classical field theories and quantum electrodynamics, this concept ensures that different representations related by gauge transformations correspond to the same physical reality [4],[6]. From an abstract viewpoint, gauge invariance provides a systematic way to separate physically meaningful dynamics from redundant coordinate or potential offsets. This symmetry-based perspective has proven essential for constructing robust and consistent models in physics, and it motivates the extension of invariance principles beyond their traditional domain toward engineered dynamical systems subject to measurement distortions and unknown offsets.

In linear control systems, similar representation issues arise in the presence of redundant measurements or sensing architectures with non-minimal output representations. In particular, when the output matrix does not have full row rank, multiple output vectors may correspond to the same underlying state. This situation naturally appears in multi-sensor configurations, distributed systems, and sparse measurement settings, where the dimension of the measurement space exceeds the effective information content [1,?]. As a result, the output representation contains directions that do not contribute to state observability but nevertheless affect the estimation process.

From this perspective, the measurement space can be decomposed into physically meaningful components and redundant directions that do not affect the underlying state. However, conventional observer designs typically treat all output components uniformly, without explicitly distinguishing between these subspaces. This may lead to estimation dynamics that are influenced by representation-dependent artifacts, rather than purely by physically relevant discrepancies.

This observation suggests that the estimation process should be constructed in a way that isolates the physically relevant part of the output from its redundant representation. A natural mechanism to achieve this separation is through a projection operator acting on the measurement space, which removes components lying in the non-informative directions while preserving the observable content. Such projection-based constructions are also well known in physics, particularly in gauge theories, where physically equivalent configurations are related by symmetry transformations and a representative is selected via gauge fixing [4].

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In this work, we propose a way of constructing a projection that acts as a static filter to recover the linear output mapping in terms of its components under given constraints. This projection is given as an alternative to the classical observer design for output redundant systems, due to the fact that no dynamical estimation is necessary to reconstruct states and disturbances.

The paper is organized as follows; section 2 gives preliminaries, section 3 ...

## 2 Preliminaries

### 2.1 The System Model

Multi output linear time invariant dynamical systems are expressed as,

$$\dot{x} = Ax + B_w w,\ y = Cx + D_w w \tag{1}$$

where $x \in \mathbb{R}^{n\times 1}$ represents the state, $w \in \mathbb{R}^{r\times 1}$ originally the bias or more generally the exogenous input and $y \in \mathbb{R}^{p\times 1}$ is the measured output. The system matrices are defined as $A \in \mathbb{R}^{n\times n}$, $B_w \in \mathbb{R}^{n\times r}$, $C \in \mathbb{R}^{p\times n}$ and $D_w \in \mathbb{R}^{p\times r}$. Output redundant systems, which is a subset of dynamical control systems given in (1), satisfy the following rank condition

$$\text{rank}(C) < p \tag{2}$$

and arise in sparse sensor applications[5], topologies used in Multi Agenr Systems(MAS)[3] and distributed networks [8]. These systems are named Strictly Output Redundant(SOR) system according to [7].

### 2.2 The Luenberger Observer

The Luenberger Observer is defined as,

$$\dot{\hat{x}} = A\hat{x} + L(y - \hat{y}),\ \hat{y} = C\hat{x} \tag{3}$$

where $L \in \mathbb{R}^{n\times p}$ is the observer gain. Assuming $(A, C)$-pair observable, the error dynamics are obtained for the error $e \triangleq x - \hat{x}$ as,

$$\dot{e} = (A - LC)e + (B_w - LD_w)w \tag{4}$$

The $\mathcal{H}_\infty$ optimal observer for objective function $z = C_z e$ is designed using the following LMI problem[2],

$$\begin{aligned} &\min_{Y,P}(\gamma) \quad \text{s.t.} \\ &\begin{bmatrix} (PA - YC) + \star & PB_w - YD_w & C_z^T \\ \star & -\gamma I & 0 \\ \star & 0 & -\gamma I \end{bmatrix} \prec 0 \\ &P \succ 0 \end{aligned} \tag{5}$$

where the observer gain is recovered with $L = P^{-1}Y$.

### 2.3 Miscellaneous

The null space of a matrix $X$ is denoted by $\mathcal{N}(X)$ where the columns space of $X$ is denoted by $\mathcal{R}(X)$. The Moore-Penrose pseudo-inverse is $(.)^\dagger$ and defined as,

$$X^\dagger = X^T(XX^T)^{-1} \tag{6}$$

## 3 Projection Filtering

In any linear dynamical system with linear output mapping where output redundancy is present, the structure can be exploited to separate the output into its components. By using Gauge-Invariance principle a projection operator can be chosen to truncate some of the subcomponents of the output mapping to isolate the component under interest.

### 3.1 The Projection

Let $X$ be a matrix and $X^\perp = \mathcal{N}(X^T)$ be its orthogonal representation, to separate the linear mapping

$$X + X^\perp \tag{7}$$

into its components, the following projection is proposed,

$$\Pi^X \triangleq I - X(X^TX)^{-1}X^T \tag{8}$$

which satisfies,

$$\Pi^X X = 0 \tag{9}$$

and

$$\Pi^X X^\perp = X^\perp \tag{10}$$

where $\Pi^X \in \mathbb{R}^{s\times s}$.

The projection operator can be applied to the measured output defined in (1) to eliminate $x$ or $w$. To eliminate $x$ the projection $\Pi^C$ is used as follows,

$$\Pi^C y = \Pi^C D_w w \tag{11}$$

and similarly, $w$ is eliminated as follows,

$$\Pi^{D_w} y = \Pi^{D_w} Cx \tag{12}$$

In (11) the projection eliminates the state dependent component of the output, isolating the exogenous component establishing state invariance. The exogenous input can be recovered under the following condition,

$$\mathcal{R}(D_w) \not\subseteq \mathcal{R}(C) \tag{13}$$

so that

$$\Pi^C D_w \neq 0 \tag{14}$$

with the least square estimator,

$$\hat{\omega} = (\Pi^C D_w)^\dagger (\Pi^C y) \tag{15}$$

In a similar manner the state can be recovered using,

$$\hat{x} = (\Pi^{D_w})^\dagger (\Pi^{D_w} y) \tag{16}$$

Thus, both the state and disturbance signals are reconstructed rather than being estimated, exploiting the structural advantage of the output redundancy of systems defined with (1). As it is illustrated in the next section, using the projection filtering, the measured output is decomposed into its state and disturbance components algebraically, which shows that the projection filtering eliminates the requirement of individual observers to estimate the states and disturbances of such systems.

## 4 Numerical Examples

### *4.1 Example 1*

The electrical system given in[7] with parameters $C_t = 2.2\text{mF}$, $L_t = 1.8\text{mH}$ and $R_t = 0.2\Omega$ is defined as the following state space model,

$$A = \begin{bmatrix} 0 & \frac{1}{C_t} \\ -\frac{1}{L_t} & -\frac{-R_t}{L_t} \end{bmatrix}, \; B_w = \begin{bmatrix} -\frac{1}{C_t} \\ 0 \end{bmatrix}, \; C = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \tag{17}$$

where output redundancy is created by replicating the sensor. The exogenous input matrix is constructed as follows,

$$D_w = C^\perp + C \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1.7071 \\ 0.2929 \\ 1.0000 \end{bmatrix} \tag{18}$$

A Luenberger observer is designed solving the LMI given in (5) and is obtained as,

$$L = \begin{bmatrix} -326.1332 & 366.9908 & -5.2903 \\ -33.4833 & 28.1920 & 48.9023 \end{bmatrix} \tag{19}$$

noting that $B_w - LD_w = 0$ is satisfied, hence the state estimation is not affected by the exogenous input. The estimated state can be used to obtain the exogenous input by utilizing,

$$\hat{w} = D_w^\dagger (y - Cx) \tag{20}$$

On the other hand, using the projection filtering the projection matrix given by (8) is calculated for $C$ as,

$$\Pi^C = \begin{bmatrix} 0.5 & -0.5 & 0 \\ -0.5 & 0.5 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{21}$$

and for $D_w$ as,

$$\Pi^{D_w} = \begin{bmatrix} 0.2714 & -0.1250 & -0.4268 \\ -0.1250 & 0.9786 & -0.0732 \\ -0.4268 & -0.0732 & 0.7500 \end{bmatrix} \tag{22}$$

The disturbance and the states are filtered using (15) and (16), respectively. The state estimation and the projection state filtering is compared in Figure 1.

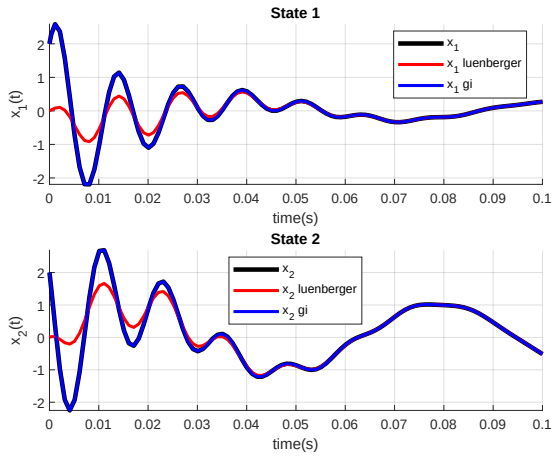


Fig. 1. State filtering versus estimation of Example 1.

The exogenous input estimated and filtered is depicted in Figure 2.

The projection filtering method utilizes the geometric properties of the output mapping, hence no dynamic estimation is present, it directly extracts the output map into its components. Despite the fact that the Luenberger observer gain for output redundant systems is able to eliminate the exogenous input term in the error dynamic by setting $B_w - LD_w$ to zero, the structure

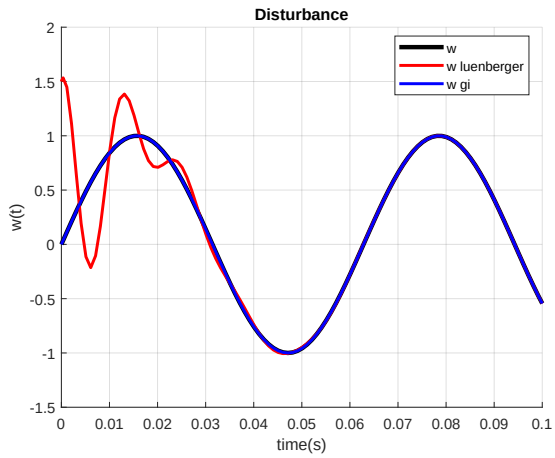


Fig. 2. Disturbance

exploit performs better for the given example since no estimation takes place. Static output filtering fails if

$$D_w = C\alpha \tag{23}$$

since,

$$\Pi y = \Pi C x + \Pi D_w w = \Pi C(x + \alpha) = 0 \tag{24}$$

which shows that different channels need to exist in order separate signals using the projection.

### *4.2 Example 2*

Let the output mapping,

$$y = Cx + D_w w + D_n n \tag{25}$$

be given as,

$$y = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} x(t) + \begin{bmatrix} 2.2060 \\ 2.3970 \\ 1.3015 \\ 1.3015 \end{bmatrix} w(t) + \begin{bmatrix} 0.1960 \\ 0.7382 \\ 0.6697 \\ 0.5265 \end{bmatrix} n(t) \tag{26}$$

where $w(t)$ is disturbance, $n(t)$ is noise and $x(t)$, which represents the system states, for illustrative purposes is chosen as,

$$x(t) = \begin{bmatrix} \sin(t) \\ \cos(t) \end{bmatrix} \tag{27}$$

The output mapping is modified as,

$$y = Cx + \begin{bmatrix} D_w & D_n \end{bmatrix} \begin{bmatrix} w \\ n \end{bmatrix} \tag{28}$$

and the projection is chosen as,

$$\Pi^{wn} = I - \begin{bmatrix} D_w & D_n \end{bmatrix} \left( \begin{bmatrix} D_w \\ D_n \end{bmatrix} \begin{bmatrix} D_w & D_n \end{bmatrix} \right) \begin{bmatrix} D_w \\ D_n \end{bmatrix} \tag{29}$$

$$\Pi^{wn} = \begin{bmatrix} 0.6224 & -0.2581 & -0.3034 & -0.2763 \\ -0.2581 & 0.1118 & 0.1698 & 0.0618 \\ -0.3034 & 0.1698 & 0.5564 & -0.3549 \\ -0.2763 & 0.0618 & -0.3549 & 0.7094 \end{bmatrix} \tag{30}$$

The state $x(t)$ is recovered using,

$$\hat{x} = (\Pi^{wn} C)^{\dagger} (\Pi^{wn} y) \tag{31}$$

The output filtering results are depicted in Figure 3.

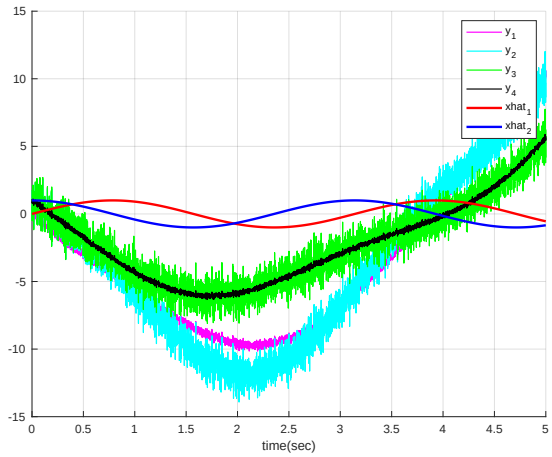


Fig. 3. State recovery

The disturbance is recovered using the projection,

$$\Pi^{xn} = I - \begin{bmatrix} C & D_n \end{bmatrix} \left( \begin{bmatrix} C \\ D_n \end{bmatrix} \begin{bmatrix} C & D_n \end{bmatrix} \right) \begin{bmatrix} C \\ D_n \end{bmatrix} \tag{32}$$

$$\Pi^{xn} = \begin{bmatrix} 0.7254 & -0.3627 & 0.1507 & 0.2120 \\ -0.3627 & 0.1813 & -0.0754 & -0.1060 \\ 0.1507 & -0.0754 & 0.0313 & 0.0440 \\ 0.2120 & -0.1060 & 0.0440 & 0.0619 \end{bmatrix} \tag{33}$$

as depicted in Figure 4. The state and disturbance recovery is successful as it can be observed from Figure 3 and Figure 4.

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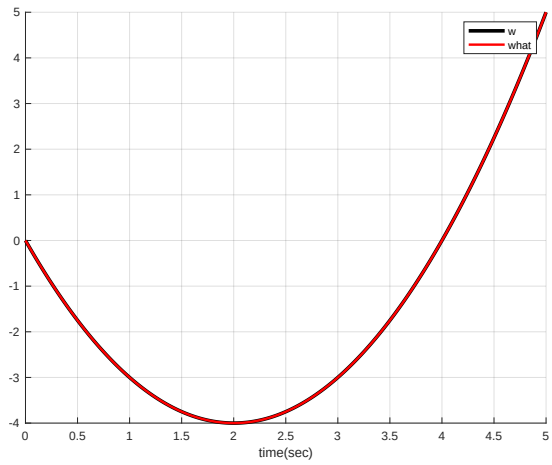


Fig. 4. Disturbance recovery

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## A A summary of Latin grammar

## B Some Latin vocabulary